

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

Use the relativistic non relativistic approximation to relate momentum and kinetic energy : $m_e c^2 \hat{=} p^2 / 2 E$ for electron motion near rest .

Given kinetic energy $E = 1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$, with $c = 2.998 \times 10^8 \text{ m/s}$, then $p = \text{sqrt}(2 m_e E/c^2) \hat{=} \text{sqrt}(2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19}) \hat{=} 5.92 \times 10^{-15} \text{ kg}\hat{\cdot}\text{m/s}$.

The de Broglie wavelength is $\hat{=} h/p$, with $h = 6.626 \times 10^{-34} \text{ J}\hat{\cdot}\text{s}$, giving $\hat{=} (6.626 \times 10^{-34})/(5.92 \times 10^{-15}) \hat{=} 1.13 \times 10^{-19} \text{ m} \hat{=} 1.23 \times 10^{-7} \text{ cm}$.

Correct answer: G. 1.23×10^{-7} cm

</think>

<answer>

G. 1.23×10^{-7} cm

</answer>

